

smallest scales—where particles behave in ways that defy classical logic. This unconventional behaviour gives quantum computers abilities that traditional systems cannot replicate.

From bits to qubits: Classical computers use bits as their basic unit of information, each representing either a 0 or a 1. Every calculation, no matter how complex, is built on combinations of these binary states. Quantum computers, in contrast, use qubits—quantum bits—that can exist as 0, 1, or both simultaneously. This phenomenon, known as superposition, allows a quantum computer to explore many possibilities at once. Imagine trying to solve a maze: a classical computer checks one path at a time, but a quantum computer can analyse all paths simultaneously, greatly accelerating certain computations.

Superposition: Many states at once: A qubit's ability to be in multiple states enables massive parallelism. For example, while two classical bits can represent one of four possible combinations at any given time, two qubits can represent all four combinations simultaneously. As the number of qubits grows, this parallel capacity expands exponentially, offering potential solutions to problems far beyond the reach of classical processors.

Entanglement: Linking qubits beyond space: Another hallmark of quantum computing is entanglement—a mysterious correlation between qubits such that the state of one instantly affects the other, no matter the distance between them. A simple analogy is a pair of magical dice that always show the same number, no matter how far apart they are rolled. This interconnectedness allows quantum computers to coordinate and process information in highly efficient ways, producing outcomes that rely on the collective behaviour of qubits rather than independent actions.

Quantum interference: Amplifying the right answers: Quantum algorithms also use interference, the ability of quantum states to reinforce or cancel out each other. This ensures that correct solutions are strengthened while incorrect ones are diminished. It's like tuning an orchestra so that only the harmonious notes resonate—leading to more accurate results.

Why it matters: Transformative applications: These unique properties aren't just theoretical marvels—they have profound real-world potential. Quantum computing could revolutionise:

- **Cryptography:** Breaking classical encryption or creating unbreakable quantum-secure codes.
- **Optimisation:** Solving logistics, scheduling, and supply chain problems faster than any classical algorithm.
- **Drug discovery:** Simulating molecules and chemical interactions with atomic precision.
- **Artificial intelligence:** Accelerating machine learning through quantum-enhanced data processing.

Quantum computers don't simply run faster—they think differently. They tackle specific classes of problems (like factoring large numbers or searching unsorted databases) exponentially faster than classical systems. While they won't replace conventional computers for all tasks, they will complement them in solving challenges that today's machines cannot feasibly handle.

In essence, what makes quantum computing unique is its departure from binary certainty into a world of probabilistic power—where information exists in multiple states, relationships transcend distance, and interference steers computation towards truth.

Understanding qubits

At the heart of quantum computing

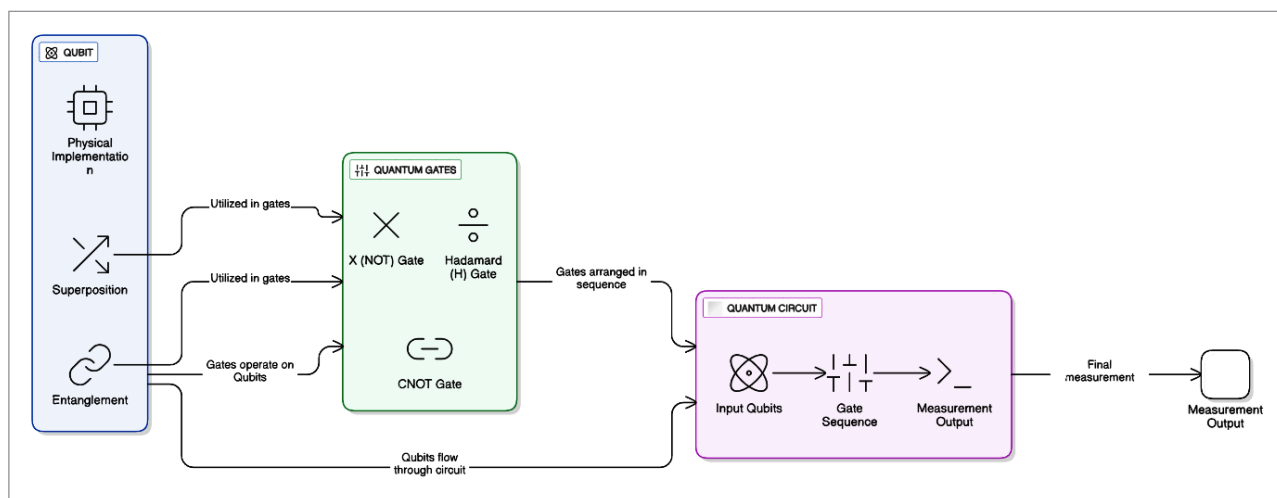


Figure 1: Reference architecture of quantum infrastructure